

Assessment Task No.1

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BSCS-3B

Machine Learning on Game-Playing AI

Chosen ML Application

Game-Playing AI – A system for self-learning AI with the capability to learn and adapt with player's stats and performance.

Problem Descriptions

Many players experiences difficulty imbalance with levels that has gameplaying AI, may it be too easy or too difficult for the players. I wish to train a model with the capability to learn and adapt with the player's performance so that it will entertain them to challenge themselves.

Part A – Identify the ML Problem

1. What is the application?

Game-Playing AI

2. What problem are you trying to solve?

The difficulty of the AI on each level and their level of adaptability.

3. What are the inputs?

Gameplay Videos, game plans through fan websites, and performance statistics.

4. What is the expected output?

The AI has the capability to adapt and learn throughout the whole game, such as imitating how the pro players play on a competition.

5. What patterns or relationships might the ML system learn

Pattern between player's skills, gameplay and the appropriate strategy on player.

6. What data would the system need?

Player's gameplay data, like their strategies, performance statistics, and their actions.

Part B – Traditional Programming vs Machine Learning

Traditional Programming

Input – Algorithm – Output

Rules/Instructions

1. When the player is losing, decrease the difficulty by lowering the stats
2. Vice versa, increase the difficulty by adding more stats to the ai.
3. If the player is using the same strategy each time, counter it so that the player has to come up with new strategy.
4. If the player has reached a certain level, increase the stats of the ai to be more challenging.

Machine Learning

Input + Output/Data → ML Algorithm

1. Player's performance statistics
2. Player's strategies
3. Player's Wins or Losses
4. Player's Gameplay
5. AI's performance with certain strategies
6. Difficulty of AI on a specific level

Part C – Create Your Own Mini Dataset

Input	Output
Frequently avoids Attack A	Use Attack B more often
Defeats weak enemies quickly	Increase Enemy Stats
Player's keep losing on a single level	Decrease level difficulty
Frequently gets hit by Attack B	Use Attack A more often
Survive the level for too long	Increase the difficulty by increasing the stats

Pattern/Relationship

How does a Machine Learning algorithm learn from data?

The performance of the player determines the difficulty of the AI on a specific level so that it's more challenge and entertaining for the player, playing the same strategies will be countered by the AI so that the player will have to use another strategy to be able to beat that level. AI can learn the relationship between the player's gameplay patterns and strategies and counter them with appropriate strategy. Base on my experience, players tend to rely on a single strategy if it always works for them.

Part D – Explain the Training Process

A Machine Learning algorithm can learn from gameplay data such as the player's win rate, score, and performance, with the AI difficulty as the output. During the training of the model, the algorithm compares the player's data with the corresponding difficulty level to identify a relationship between player skill and AI difficulty. The difficulty will be

adjusting base on the performance of the player, if the player plays well, it will increase the difficulty by using different strategies to counter the player and increase the stats of the AI, vice versa with the poor performance of the player, it will decrease the difficulty by lowering the stats of the AI and use the generic strategies. After training, the AI can predict the suitable difficulty for players with good and poor performance so that it will be more entertaining and challenging.

Reflections

Why is learning from data useful instead of manually programming every possible rules?

Machine Learning has the capability to learn and adapt base on the data it receives, and the more data the it receives the better it predicts and it can decision on their own. Unlike the traditional programming, the programmer has to think of every possible rules so that it produces the desirable outcome to the user. In summary, machine learning works by learning through patterns from data to predict the output, and through training the model, the accuracy of the predictions will become more precise if it's been exposed to many datasets.